



Text Analytics

Introduction

2019–20

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<http://www.aueb.gr/users/ion/>

Contents

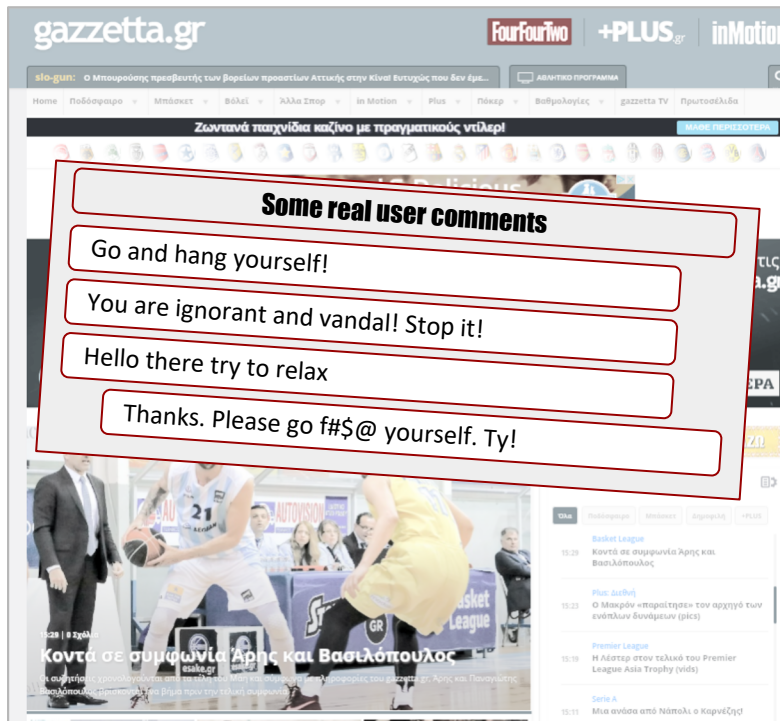
- **What** this course is about.
- **Examples** of applications.
- **Topics** to be covered.
- **Organization** of the course:
 - Lectures, demos.
 - Assignments, exams, grading policy.

Opinion mining

Apple Mac mini	GO
money, price, cost, ...	☆☆☆☆☆
ram, memory, ...	☆☆☆
design, color, feeling, ...	☆☆☆☆
extras, keyboard, screen, ...	☆☆

- From **customer reviews** (<http://alt.qcri.org/semeval2016/task5/>).
- From **tweets** (<http://alt.qcri.org/semeval2016/task4/>), **FB updates**, **blogs**, **newspapers**, **radio/TV broadcasts**, ...

User comment moderation



The screenshot shows the Gazzetta.gr website with a basketball article. A red-bordered box highlights several user comments:

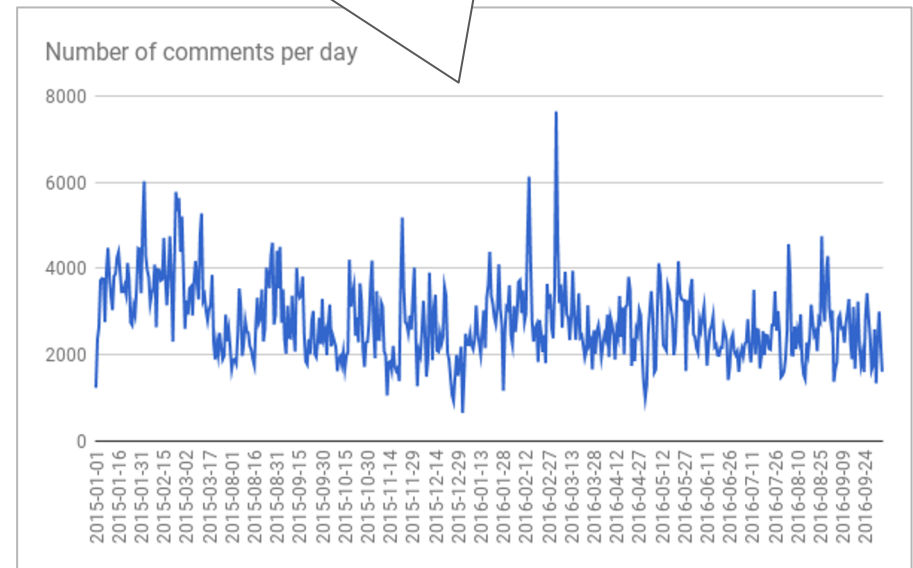
Some real user comments

- Go and hang yourself!
- You are ignorant and vandal! Stop it!
- Hello there try to relax
- Thanks. Please go f#\$@ yourself. Ty!

Moderators to reject abusive comments, avoid reputational damage, fines, putting off readers...



Thousands of comments per day require moderation. Some very disturbing...



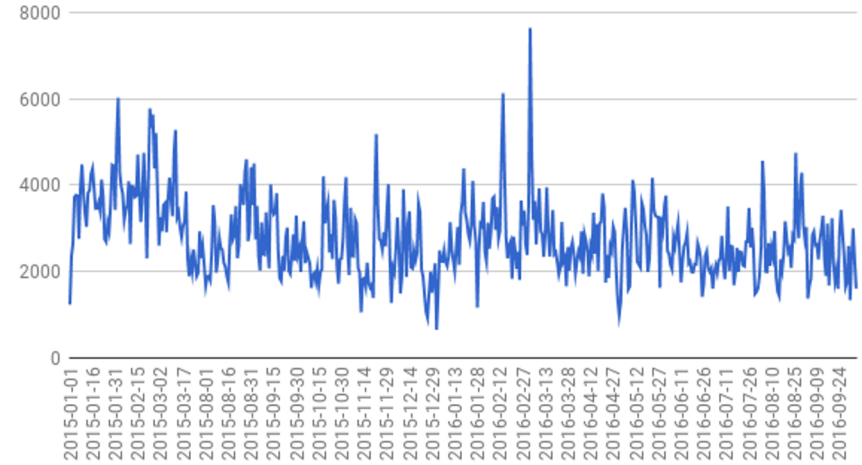
J. Pavlopoulos, P. Malakasiotis and I. Androutsopoulos, “Deeper Attention to Abusive User Content Moderation”, EMNLP 2017, <http://nlp.cs.aueb.gr/pubs/emnlp2017.pdf>.

User comment moderation

A moderation panel assists the moderators to detect abusive comments, and leads to quicker publication of non-abusive comments.

Highlighting suspicious words using an RNN with self-attention.

Number of comments per day



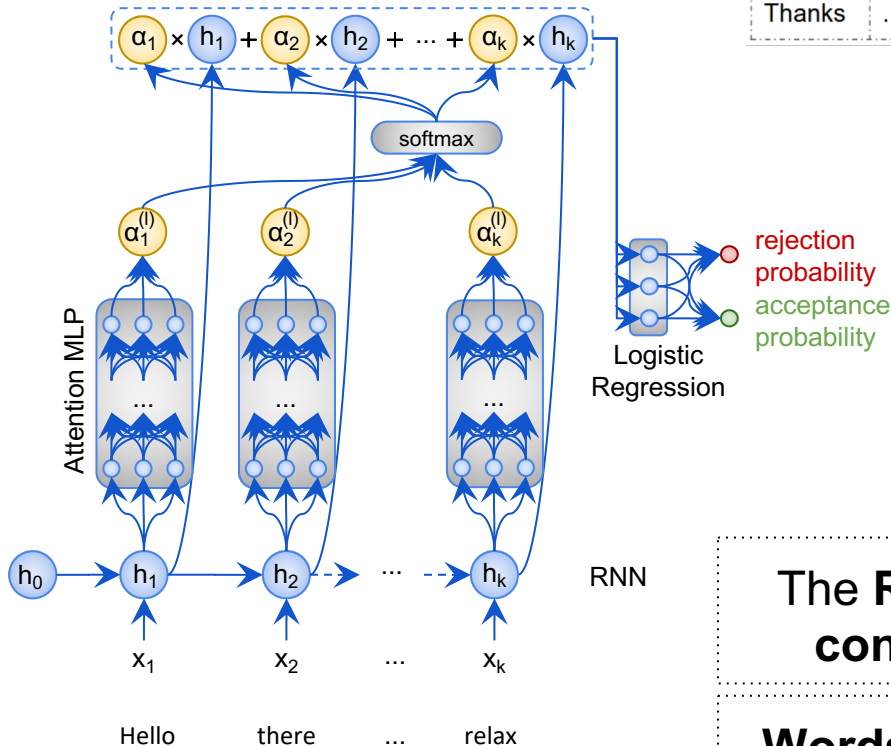
Moderation Panel

Go	and	hang	yourself	!						85%		
You	are	ignorant	and	vandal	!	Stop	it	!		88%		
Hello	there	try	to	relax						0%		
Thanks	.	Please	go	f#\$@	yourself	.	Ty	!		85%		

RNN with deep self-attention (α -RNN)

The **attention scores** can also be used to **highlight suspicious words!**

Go	and	hang	yourself	!				
You	are	ignorant	and	vandal	!	Stop	it	!
Thanks	.	Please	go	fuck	yourself	.	ty	!



An **LR layer** uses the **attention-weighted sum** of the **RNN states** as a feature vector.

Each state of the RNN is used as **input** to a **self-attention MLP** (4 layers, same MLP at each time step) that scores the **importance** of the corresponding **word**.

The **RNN states** make the **word embeddings context-aware**. We use **GRUs** (128 dims).

Words are mapped to **embeddings** (word2vec, 300 dimensions).

J. Pavlopoulos, P. Malakasiotis and I. Androutsopoulos,, “Deeper Attention to Abusive User Content Moderation”, EMNLP 2017, <http://nlp.cs.aueb.gr/pubs/emnlp2017.pdf>.

Extracting Contract Elements

THIS AGREEMENT is made the 15th day of October 2009
(The “Effective Date”) BETWEEN:

- (1) **Sugar 13 Inc.**, a corporation whose office is at James House,
42-50 Bond Street, London, EW2H TL (“**Sugar**”);
- (2) **E2 UK Limited**, a limited company whose registered office is
at 260 Bathurst Road, Yorkshire, SL3 4SA (“**Provider**”).

RECITALS:

- A. The Parties wish to enter into a framework agreement which
will enable Sugar, from time to time, to [...]
- B. [...]

NO THEREFORE IT IS AGREED AS FOLLOWS:

ARTICLE I - DEFINITIONS

- “**Sugar**” shall mean: Sugar 13 Inc.
- “**Provider**” shall mean: E2 UK Limited
- “**1933 Act**” shall mean: Securities Act of 1933

ARTICLE II - TERMINATION

The Service Period will be for five (5) years from the Effective Date
(The “**Initial Term**”). The agreement is considered to be terminated
in October 16, 2014.

ARTICLE III - PAYMENT - FEES

During the service period monthly payments should occur. The
estimated fees for the Initial Term are £154,800.

ARTICLE IV - GOVERNING LAW

This agreement shall be governed and construed in accordance with
the Laws of England & Wales. Each party hereby irrevocably
submits to the exclusive jurisdiction of the courts sitting in Northern
London.

IN WITNESS WHEREOF, the parties have caused their
respective duly authorized officers to execute this Agreement.

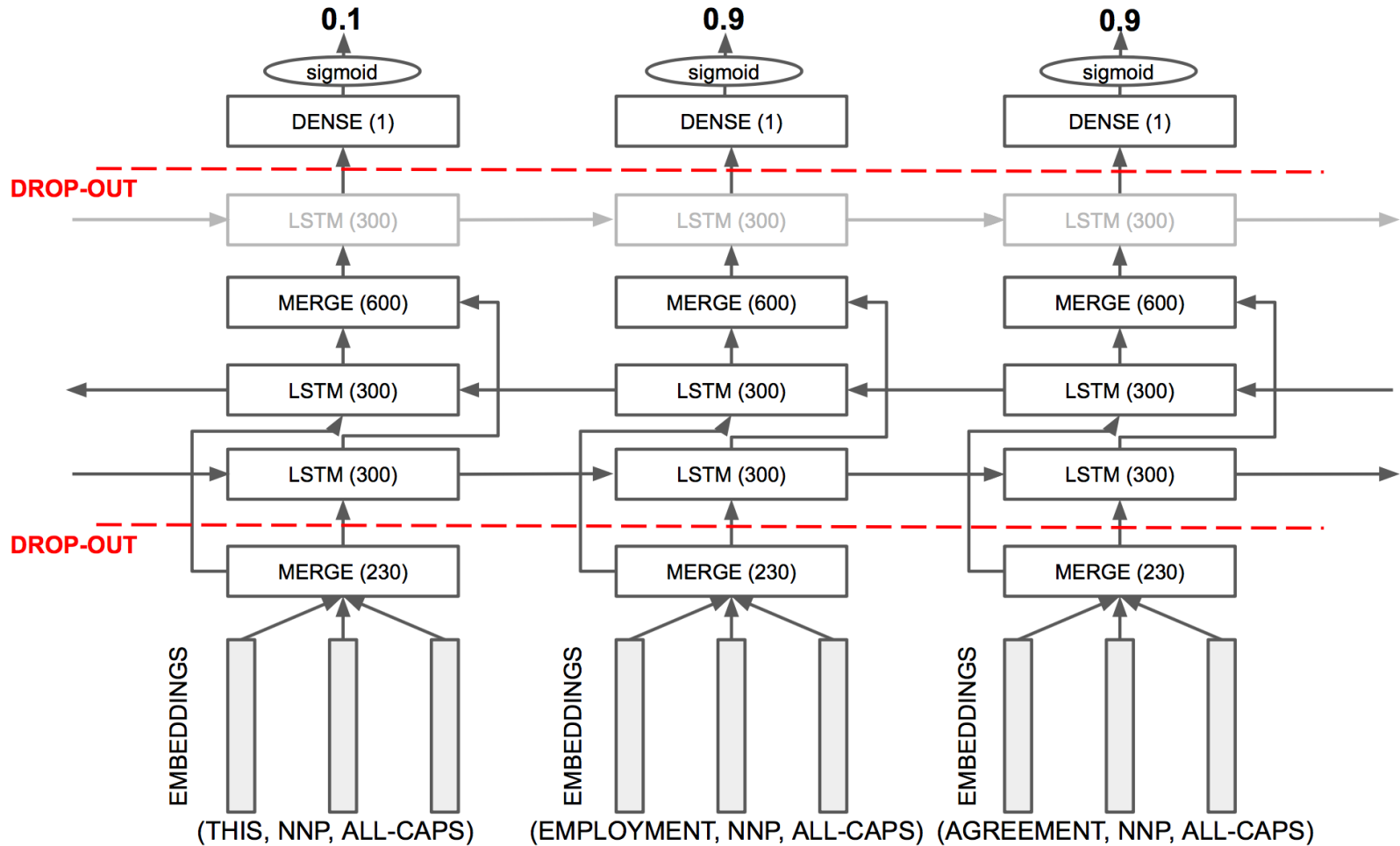
BY: George Fake
Authorized Officer
Sugar 13 Inc.

BY: Olivier Giroux
CEO
E2 UK LIMITED

Extract start/end dates,
durations, contractors,
amount, legal references,
jurisdiction etc.

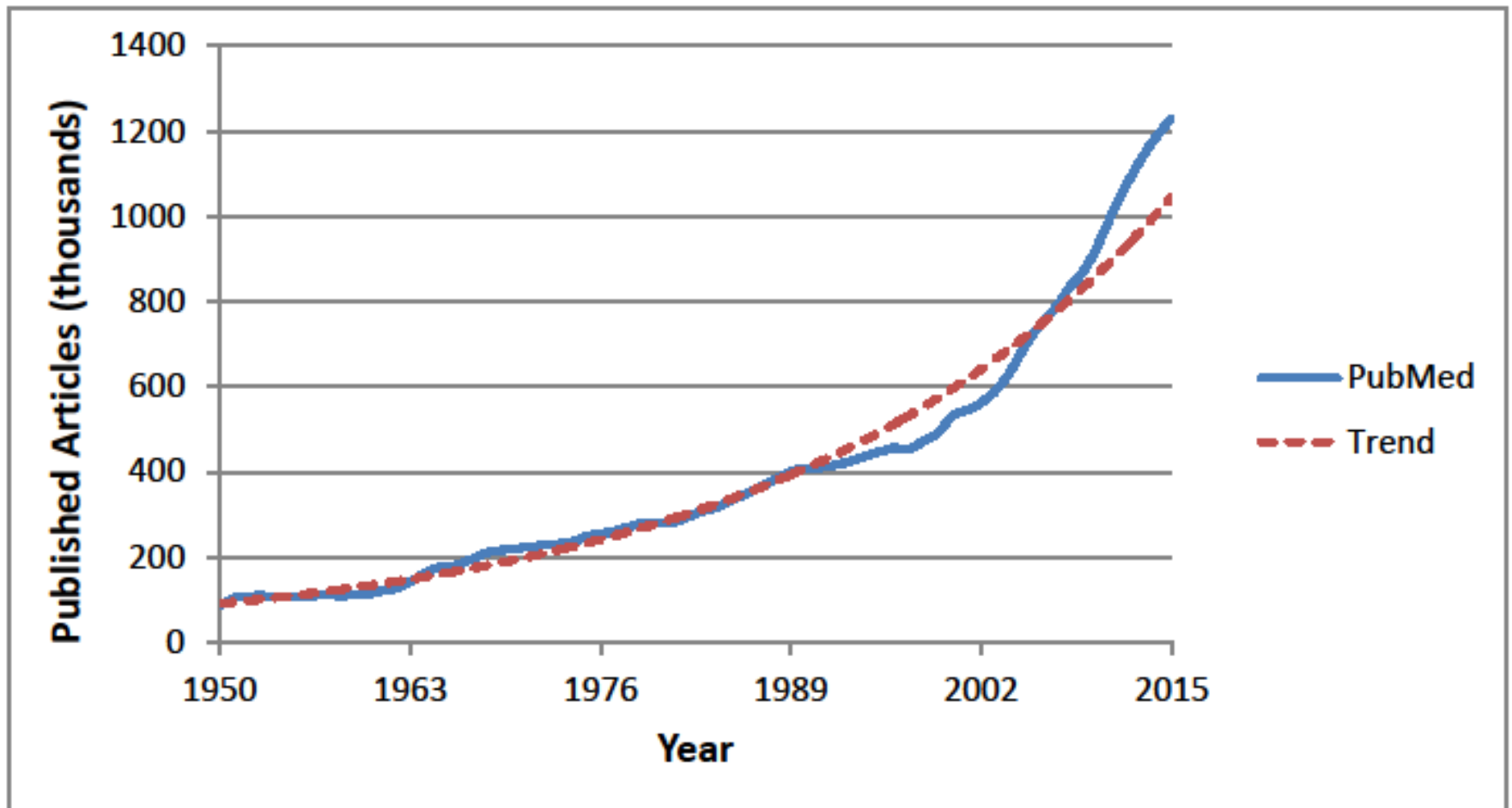
- I. Chalkidis, I. Androutsopoulos and A. Michos, “Extracting Contract Elements”, ICAIL 2017,
<http://nlp.cs.aueb.gr/pubs/icail2017.pdf>.
- I. Chalkidis and I. Androutsopoulos, “A Deep Learning Approach to Contract Element
Extraction”, JURIX 2017, <http://nlp.cs.aueb.gr/pubs/jurix2017.pdf>.

Extracting Contract Elements



I. Chalkidis and I. Androutsopoulos, “A Deep Learning Approach to Contract Element Extraction”, JURIX 2017, <http://nlp.cs.aueb.gr/pubs/jurix2017.pdf>.

Biomedical articles per year



Questions of biomedical experts

Factoid question

“What is the methyl donor of DNA (cytosine-5)-methyltransferases?”

Exact answer

“S-adenosyl-L-methionine”

Summary answer

“S-adenosyl-L-methionine (AdoMet, SAM) is the methyl donor of DNA (cytosine-5)-methyltransferases. DNA (cytosine-5)-methyltransferases catalyze the transfer of a methyl group from S-adenosyl-L-methionine to the C-5 position of cytosine residues in DNA.”

Questions of biomedical experts

List question

“Which species may be used for the biotechnological production of itaconic acid?”

Exact answer

“Aspergillus terreus”, “Aspergillus niger”, “Ustilago maydis”

Summary answer

“In 1955, the production of itaconic acid was firstly described for Ustilago maydis. Some Aspergillus species, like A. itaconicus and A. terreus, show the ability to synthesize this organic acid and A. terreus can secrete significant amounts to the media. Itaconic acid is mainly supplied by biotechnological processes with the fungus Aspergillus terreus. Cloning of the cadA gene into the citric acid producing fungus A. niger showed that it is possible to produce itaconic acid also in a different host organism.”

Selecting relevant snippets

Release of sunflower seed dormancy by cyanide: cross-talk with ethylene signalling pathway

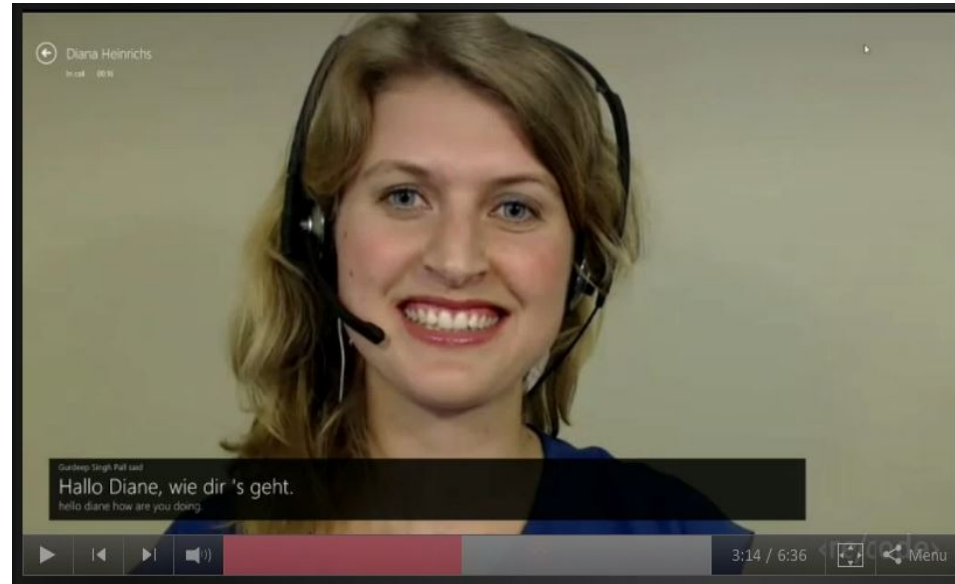
Introduction Cyanide is a compound known to stimulate germination and to release dormancy of seeds of many species (Taylorson and Hendricks, 1973; Roberts and Smith, 1977; Bogatek and Lewak, 1988; Côme et al., 1988; Bethke et al., 2006). Seed dormancy is defined as the property of a seed that prevents its germination in apparently favourable conditions (Finch-Savage and Leubner-Metzger, 2006). Despite the well-described effects of cyanide on germination and dormancy, the cellular bases of its mechanism are poorly understood, and seem moreover to vary from one species to another. Different hypotheses have been proposed to explain the stimulatory effect of cyanide on germination and dormancy (Côme and Corbineau, 1989). ✕ According to Taylorson and Hendricks (1973), the cyanhydric gas could react with L-cysteine to give the β -L-cyanoalanine necessary for the synthesis of arginine and aspartic acids which could be limiting factors for germination. Hagesawa et al. (1994) suggested that this increase in the amino acid pool might also promote germination by decreasing the water potential in embryonic axis. However, other respiratory inhibitors which are not metabolized, such as NaN_3 or Na_2S , have the same effect as KCN in various species (Roberts and Smith, 1977; Côme and Corbineau, 1989). Some studies proposed that the beneficial effect of cyanide on germination might involve the cyanide-insensitive pathway (Esashi et al., 1979, 1981b; Upadhyaya et al., 1983), the pentose phosphate pathway (Roberts and Smith, 1977; Côme and Corbineau, 1989), the glycolysis (Bogatek, 1995) or the hydrolysis of oligosaccharides and their catabolism (Bogatek and Lewak, 1991; Bogatek et al., 1999). Cyanide is also known to interact with reactive oxygen species (ROS) metabolism; it is an inhibitor of Cu/Zn superoxide dismutase (SOD) (Bowler et al., 1992) and catalase (CAT) (Tejera García et al., 2007) and it has been demonstrated to induce oxidative stress and lipid peroxidation in animals (Johnson et al., 1987; Gunasekar et al., 1998). Oracz et al. (2007) recently demonstrated that cyanide could trigger protein oxidation during sunflower seed dormancy alleviation. At last, cyanide might also interplay with the ethylene signalling pathway. Indeed, hydrogen cyanide is a co-product of ACC oxidase, which converts ACC to ethylene (Peiser et al., 1984), and it has been proposed to stimulate ethylene biosynthesis via a feedback effect (Pirrung and Brauman, 1987). Thus Smith and Arteca (2000) demonstrated that the ACC synthase gene ACS6 was activated by cyanide in Arabidopsis. However, it is actually not known whether ethylene and cyanide share some molecular components of their downstream transduction pathways. The putative relationship between cyanide and ethylene signalling pathways might be particularly relevant for sunflower seeds, whose dormancy is broken by ethylene (Corbineau et al., 1990). The inability of freshly harvested sunflower seeds to germinate at temperatures below c. 15 °C results from an embryo dormancy which is gradually eliminated during dry storage (Corbineau et al., 1990; Corbineau and Côme, 2003). Embryo dormancy is characterized by the inhibition of radicle extension thus preventing excised embryos to grow (Finch-Savage and Leubner-Metzger, 2006). Little attention has been paid to the

Information from structured data

List question: *“Which forms of cancer is the Tpl2 gene associated with?”*

- Related **RDF triple**:
 - Subject: <http://www4.wiwiss.fu-berlin.de/diseasome/resource/diseases/3003> (lung cancer)
 - Predicate: <http://www4.wiwiss.fu-berlin.de/diseasome/resource/diseasome/associatedGene>
 - Object: <http://www4.wiwiss.fu-berlin.de/diseasome/resource/genes/TPL2>
- Related **concepts**:
 - <http://www.disease-ontology.org/api/metadata/DROID:162> (cancer)
 - http://www.uniprot.org/uniprot/M3K8_RAT (TPL2 synonym)

Machine translation



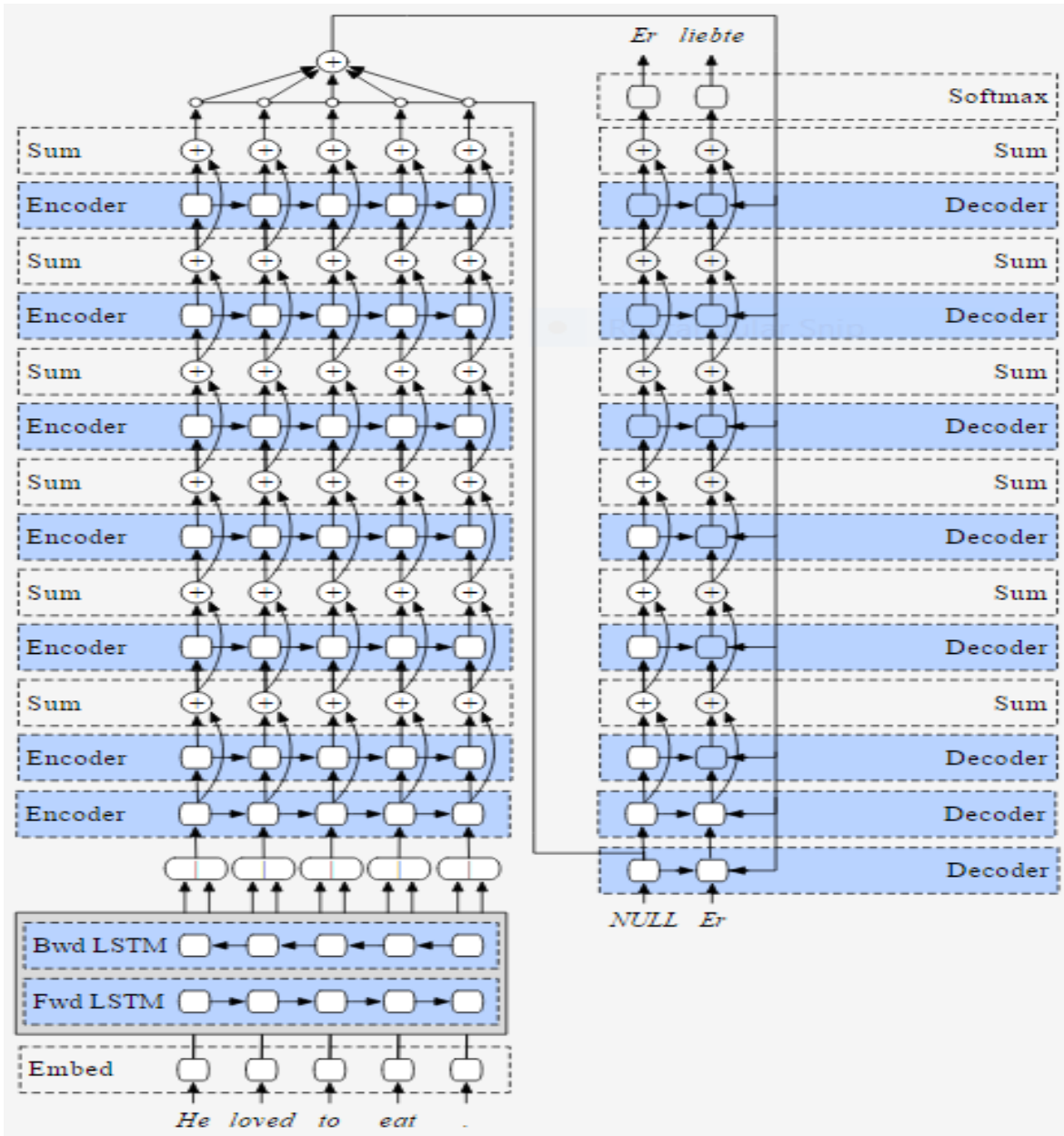
Watch the videos:

<http://bit.ly/2XEyX0t>

<https://www.youtube.com/watch?v=tRG55fTLluU>

<http://t.co/waThtqdbMO>

Google's RNN-based MT system



Google's paper:
<https://arxiv.org/abs/1609.08144>

Images from Stephen Merity's
http://smerity.com/articles/2016/google_nmt_arch.html

Step by step
explanation in
the course...

Natural language generation



Touring Machine (G. Karakatsiotis and V. Pterneas, Dept. of Informatics, AUEB). Won the Greek Imagine Cup of Microsoft in 2010 and 1st prize in interoperability at the International Imagine Cup of 2011. The texts are generated from an OWL ontology of the Ancient Agora of Athens, which was created during IST INDIGO (www.ics.forth.gr/indigo/) from information provided by the Foundation of the Hellenic World (www.ime.gr).

Language technology examples

- Algorithms, models, systems to **analyze** and **generate** **written** or **spoken natural language**.
 - **Spelling/syntax checking**, smart **keyboards**.
 - **Machine translation** of texts or speech.
 - **Filtering/routing** (e.g., spam filters, call centers).
 - **Information extraction** and **opinion mining**.
 - **Question answering** (e.g., biomedical QA).
 - **Spoken dialogue systems** and **chatbots** (e.g., booking tickets, requesting information, in-car dialogues).
 - **Text generation** (e.g., describing products in multiple languages, generating reports from sensor data).
 - **Text summarization** (e.g., of multiple retrieved docs).

Many related terms

- **Natural Language Processing (NLP):**
 - Usually considered a sub-field of **Artificial Intelligence**.
 - **Algorithms, models, systems** to “understand” or **generate mostly written** natural language.
- **Computational Linguistics:**
 - Also mostly about **written** NL, originally more emphasis on computational *linguistic* theories, but now in effect a **synonym** of NLP, and (like NLP) probably closer to CS than linguistics.
 - The *Association for Computational Linguistics* (ACL) organizes the top NLP conferences and publishes the top journals.
- **(Human) Language Technology/Engineering:**
 - Less established term, often **includes speech processing**, more emphasis on **systems/engineering**.
- **Text (and Speech) Analytics:**
 - Business term, usually meaning information extraction, sentiment analysis, opinion mining.

Course units (~1 unit/week)

- **U1:** Introduction, n-gram language models, spelling correction, text normalization.
- **U2&3:** Text classification with (mostly) linear classifiers.
- **U4&5:** Text classification with MLPs.
- **U6&7:** NLP with RNNs.
- **U8&9:** NLP with CNNs and Transformers.
- **U10:** Parsing and relation extraction.

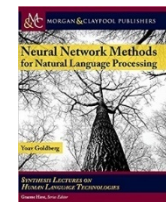
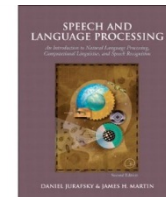
Lectures, demos, assignments

- **10 lectures/demos** (3 hours/week):
 - **Lectures** (key ideas, theory). Slides, study exercises (most solved), references provided.
 - **Demos**: in ~5 weeks, the **last hour** will be a mini-lab of related tools (e.g., NLTK, spaCy, scikit-learn, Keras, PyTorch). Code examples provided.
- **Assignments**:
 - Some of the unsolved **exercises** (probably 5, some requiring **programming**) to be handed in (**45%**).
- **Participation and final exam**: 10% + 45%

Recommended books

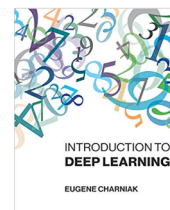
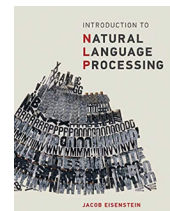
- **Main textbooks (optional):**

- *Speech and Language Processing* by D. Jurafsky and J.H. Martin, 2ⁿ edition, Pearson, 2009. (Available at AUEB's library. Draft 3rd edition: <http://web.stanford.edu/~jurafsky/slp3>)
- *Neural Network Models for Natural Language Processing*, by Y. Goldberg, Morgan & Claypool, 2017. (Available at AUEB's library.)



- **Other recommended textbooks (optional):**

- *Introduction to Natural Language Processing* by J. Eisenstein, MIT Press, 2019. (Free draft: <https://github.com/jacobeisenstein/gt-nlp-class/blob/master/notes/eisenstein-nlp-notes.pdf>)
- *Introduction to Deep Learning* by E. Charniak, MIT Press, 2019. (Available at AUEB's library.)



Recommended books

- **Older but still useful textbooks (optional):**
 - *Foundations of Statistical Natural Language Processing* by C.D. Manning and H. Schutze, MIT Press, 1999. (Available at AUEB's library.)
 - *An Introduction to Information Retrieval* by C.D. Manning, P. Raghavan and H. Schütze, Cambridge University Press, 2008. (Freely available from: <http://nlp.stanford.edu/IR-book/information-retrieval-book.html>.)

